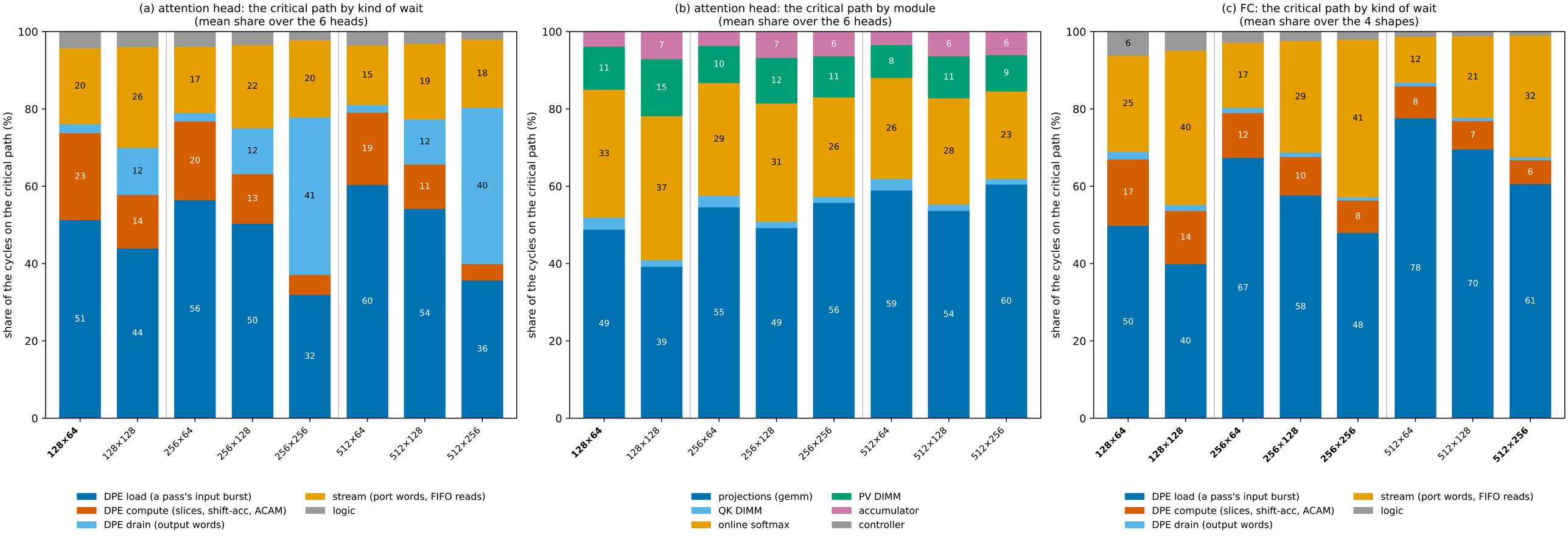
(a) attention head: the critical path by kind of wait
(mean share over the 6 heads)
share of the cycles on the critical path (%)
128×64
128×128
256×64
256×128
256×256
512×64
512×128
512×256
DPE load (a pass's input burst)
DPE compute (slices, shift-acc, ACAM)
DPE drain (output words)
stream (port words, FIFO reads)
logic
(b) attention head: the critical path by module
(mean share over the 6 heads)
projections (gemm)
QK DIMM
online softmax
PV DIMM
accumulator
controller
(c) FC: the critical path by kind of wait
(mean share over the 4 shapes)